

Composition Pair B1.04 + B1.08 — Aspect Coordination Governance Requires Modular Cells, and Cell Modularity Is Architecturally Motivated by Aspect Coordination Requirements

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize, in composition-pair structure, the mutual dependency between aspect coordination governance (B1.04) and cell modularity (B1.08) — establishing that neither commitment is fully operative without the other, and that their co-presence is what makes aspect-level coordination governance tractable and cell modularity architecturally purposeful.

Abstract

The Coordination Knowledge Substrate (CKS) pattern at the Paper 2 scope commits to two architectural properties that are individually named and individually motivated: aspects as coordination arrangements of cells organized for a purpose (B1.04), and cells as independently composable, governable, and testable units with clear operational boundaries (B1.08). This note formalizes the reinforcing relationship between them. B1.04 requires B1.08 for effective operation: aspect coordination governance organizes cell membership, invocation, and output integration, but each of these governance dimensions becomes imprecise or unpredictable if cells lack the modularity B1.08 specifies. B1.08 in turn is architecturally motivated by B1.04: cells are modular not as an abstract structural virtue but because they must be composable into aspects — the governance needs of aspect coordination define what modularity concretely requires for cells. The pair’s operational embodiment is composition configurability (B2.36): aspect membership can be reconfigured precisely because cells are independently composable, and cells are independently composable precisely because aspects need to reconfigure their members. The co-presence requirement runs in both directions: B1.04 without B1.08 produces coordination governance that is architecturally real but operationally impaired; B1.08 without B1.04 produces modular cells that satisfy a structural requirement without a coordination purpose to satisfy it in.

1. Why this pair requires explicit formalization

B1.04 and B1.08 are defined at separate points in the source paper’s architecture and can be read independently: aspects are coordination arrangements (§5.2), cells are modular units (§5.2). Neither

definition asserts a dependency on the other. Taken separately, an implementer might reason that aspects are a coordination architecture that happens to use cells, or that cell modularity is a structural property that aspects happen to benefit from — treating the connection as incidental rather than constitutive.

The connection is constitutive. The four governance functions aspect coordination performs — organizing cell membership, specifying invocation rules, integrating cell outputs, and handling coordination conflicts — each depend on a structural property of cells that only B1.08 supplies. Without clear operational boundaries, membership changes produce unpredictable outcomes. Without well-specified input and output types, invocation rules cannot be precisely stated. Without non-overlapping operational territories, conflict handling encounters conflicts that the coordination rules cannot cleanly address. Without independent testability, aspect-level verification cannot confirm that coordination rules function as designed.

The dependency runs equally in the other direction. Cell modularity, examined without B1.04's context, is an abstract structural requirement: cells should be independently composable, governable, and testable. The question “independently composable with respect to what?” has no answer without the aspect coordination context. B1.04 supplies that answer: cells are composable with respect to aspect membership — they can be added to or removed from aspects without implicit dependencies on other member cells triggering redesign. The composability requirement, the independent testability requirement, the clear interface requirement, and the independent governability requirement each gain operational content from the coordination functions aspects need to perform.

This note formalizes the reinforcing pair to make the dependency explicit, stated, and checkable.

2. The two commitments

B1.04 — Aspect as coordination arrangement. An aspect is a coordination arrangement of cells organized for a purpose. Aspect coordination governance organizes three functions: membership governance (which cells participate in the aspect, and how membership changes over time); invocation rules (how cells within the aspect are invoked for coordination, under what conditions, in what sequence or parallel arrangement); and output integration (how the outputs of member cells are combined into aspect-level results). Coordination rules per B2.16 are the substrate artifacts that carry these governance functions as human-authored, addressable content. Aspects can be decomposed into their membership configuration (B2.08), their coordination rules (B2.16), and their inheritance verification (B2.19); they can participate in multi-aspect arrangements (B2.18) and can be reconfigured through directed selection (B1.14). The central commitment is that coordination governance is an explicit, substrate-carried, human-governed architecture — not an implicit property of how cells happen to be assembled.

B1.08 — Cell modularity. A cell is an independently composable, governable, and testable unit with clear operational boundaries. The modularity commitment specifies four properties: composability (the

cell can be added to or removed from aspect membership without implicit dependencies on other cells requiring redesign); independent testability (the cell's behavior can be tested independently of its current membership configuration, enabling the determinism test per A5.06 to be run at cell scope without requiring aspect-level context); clear interface (the cell's input types and output types are well-specified, enabling invocation rules to reference them precisely and output integration rules to combine results predictably); and independent governability (the cell's DNA layer per B1.03 can be modified through directed selection per B1.14 without affecting peer cells whose operational scope is distinct). The modularity commitment is architectural: it is a requirement at instantiation, not a property hoped for through disciplined implementation.

3. B1.04 requires B1.08: how non-modular cells impair coordination governance

Aspect coordination governance operates over cells as its governance objects. The quality of governance is bounded by the structural properties of those objects. Four governance dimensions exhibit this dependency concretely.

Membership governance. Aspects add and remove cell members through composition configurability per B2.36. This membership reconfiguration presupposes that cells are independently composable: adding a new cell member does not create implicit dependencies between it and existing members; removing a cell member does not leave remaining members in a state that assumes the removed cell's presence. When cells are not modular — when they have implicit dependencies on co-members, or when their operational scope extends beyond what their substrate content specifies — membership changes become unpredictable. The undifferentiated cell anti-pattern (B3.04) is the structural failure mode here: a cell whose operational scope is undefined has no principled separation from cells around it, making membership changes in the aspect that contains it into operations with uncertain effects. Aspect membership governance cannot function predictably over cells that are not independently composable.

Conflict handling. Aspect coordination rules per B2.16 specify how conflicts between cell outputs are handled. This conflict-handling specification presupposes that cell operational territories are either non-overlapping or that overlaps are explicit. Modular cells with clear operational boundaries produce conflicts that are identifiable: two cells produce outputs that contradict, and the contradiction is nameable because each cell's operational scope is specified. Non-modular cells with vague operational territories produce conflicts that cannot be cleanly addressed because it is not clear which cell's output should prevail, whether the conflict reflects overlapping scope or genuine contradictory evidence, or how a coordination rule should partition resolution authority between the two cells. Aspect coordination rules can address identifiable conflicts; they cannot address conflicts whose structure is obscured by the operational vagueness of the cells producing them.

Invocation rules. Aspect invocation rules specify how cells are invoked: under what conditions, with what inputs, in what relationship to other member cells. This specification requires that cell input types

are well-specified — that the invocation rule can reference concrete input categories when specifying when and how a cell is invoked. When cell input types are not specified (a modularity failure per B1.08), invocation rules can be written only loosely, referencing approximate rather than precise input categories. Loose invocation rules produce governance that is nominally operative but practically imprecise: it does not cleanly determine when a cell should be invoked, and implementations of the rule by different actors at different times may diverge in ways that undermine the coordination governance B1.04 commits to.

Aspect-level verification. Aspect-level inheritance verification per B2.19 confirms that coordination rules function correctly with current cell membership. This verification requires that cell behaviors can be independently assessed — that a test can be run on a cell in isolation, its outputs confirmed against specification, and the cell’s contribution to the aspect confirmed independently of other cells’ contributions. Non-modular cells resist this independent testing: a cell whose behavior depends on the state of co-member cells, or whose outputs are entangled with the outputs of cells around it, cannot be independently assessed. Aspect-level verification then becomes a system-level test of the full membership configuration rather than a compositional verification of independently confirmed members. The verification loses the modularity that would make it tractable at scale.

4. B1.08 is motivated by B1.04: how coordination context defines what modularity requires

Cell modularity is not a structural virtue that cells should possess for its own sake. It is an architectural requirement whose content is defined by the coordination context in which cells participate. B1.04 provides that context.

Composability as coordination-membership requirement. The composability property of modularity — that a cell can be added to or removed from membership without triggering redesign — is specifically the property aspects need in order to reconfigure their membership per B2.36. Without the aspect coordination context, composability is an abstract desideratum: “cells should not have hidden dependencies.” With B1.04, composability is a concrete requirement: a cell must be able to participate in an aspect’s coordination arrangement without its operation depending on which other cells are co-members. The coordination context defines what “independently composable” means operationally.

Independent testability as coordination-verification requirement. The independent testability property — that a cell’s behavior can be tested without aspect-level context — is specifically the property that aspect-level inheritance verification per B2.19 requires. Without B1.04, independent testability is a development practice: test cells in isolation to simplify testing. With B1.04, independent testability is an architectural requirement: aspect-level governance depends on the ability to independently confirm cell behavior as a precondition for confirming that coordination rules over those cells function correctly. The verification structure of aspect governance motivates the independent testability requirement.

Clear interface as invocation-rule requirement. The clear interface property — that cell input and output types are well-specified — is specifically the property that invocation rules and output integration rules require in order to be precisely stated. Without B1.04, a clear interface is good practice. With B1.04, it is what makes the coordination governance rules in B2.16 operable: invocation rules reference cell input types, and output integration rules reference cell output types. Vague interfaces make governance rules vague. The coordination commitment defines what a cell's interface must specify.

Independent governability as directed-selection requirement. The independent governability property — that a cell's DNA layer can be modified through directed selection per B1.14 without affecting peer cells — is specifically the property that aspect-level membership governance requires. When an aspect's coordination governance changes a member cell through directed selection, it must be able to do so without inadvertently changing peer cells whose operational scope is distinct. The governance architecture of aspects is what gives independent governability its architectural standing as a cell requirement, rather than its standing as a general software property.

5. Composition configurability as pair embodiment

Composition configurability per B2.36 is the operational expression in which the reinforcing pair manifests most directly. The assessment asks: can aspect membership be reconfigured? The question has two parts, each corresponding to one commitment in the pair.

The first part — can membership be reconfigured? — is a B1.04 question: does the aspect's coordination architecture support membership change through governance? Does the aspect carry orchestration rules that specify how new cells are inducted into membership, how cell removal changes the coordination arrangement, and how coordination rules are updated when membership changes?

The second part — without redesigning cells? — is a B1.08 question: are cells independently composable? Does a cell's operation not depend on which co-members it currently has? Does removing a cell from one aspect leave it available for membership in another aspect without modification?

Both conditions must hold for composition configurability to be confirmed. B1.04 provides the governance architecture that makes membership change principled; B1.08 provides the cell properties that make membership change operationally tractable. Neither alone is sufficient.

Multi-aspect participation per B2.18 is the further expression of this pair. A single cell can participate in multiple aspects simultaneously precisely because B1.08 modularity ensures the cell's operation does not depend on which aspect it is operating within at a given moment. B1.04 provides the coordination context in which this participation is governed — each aspect in which the cell participates has its own coordination rules governing how that cell is invoked, how its outputs are integrated, and how its membership in that aspect is managed. Multi-aspect participation is possible

because B1.08 ensures the cell's independence; it is meaningful because B1.04 ensures that each aspect's governance of that cell's participation is precise.

The same pair appears in vertical evolution per B1.16. Vertical evolution may introduce new modular cells into aspects based on action-feedback evidence. The architectural sequence is: evidence accumulates that a new cell with distinct operational scope would serve an aspect's coordination purpose (B1.04 motivation); the new cell is designed as an independently composable, governable, and testable unit with clear interface (B1.08 specification); it is inducted into aspect membership through composition configurability (B2.36 operation). The pair's reinforcing structure is what makes vertical evolution tractable: new cells can be introduced into aspects without redesigning existing members because modularity is an architectural requirement, not a post-hoc structural property.

6. Co-presence requirement and failure modes

B1.04 without B1.08 — coordination governance over non-modular cells. An implementation that commits to aspect coordination governance without committing to cell modularity produces a governance architecture that is structurally real but operationally impaired. Coordination rules exist and are substrate-carried; membership governance mechanisms exist. But the rules are imprecise because cell interfaces are unspecified. Membership changes are unpredictable because cells have implicit co-member dependencies. Conflict handling is degraded because cell operational territories are vague. Verification is undermined because cells cannot be independently tested.

The Monolithic Cell anti-pattern (B3.09) captures the structural failure mode this produces. A monolithic cell — one whose operational scope spans multiple domains, whose outputs cannot be attributed to distinct operational territories, and whose behavior depends on the state of surrounding cells — is the structural outcome when B1.08 is absent. Aspects containing monolithic cells carry governance rules that nominally govern coordination but cannot be precisely applied: the cell's operational vagueness travels through the governance rules that reference it.

B1.08 without B1.04 — modular cells without coordination purpose. An implementation that commits to cell modularity without committing to aspect coordination governance produces cells that satisfy a structural requirement without an operational context in which that requirement pays off. Cells are independently composable but there is no coordination architecture that takes advantage of their composability. Cells have clear interfaces but there are no coordination rules that reference those interfaces. Cells are independently governable but there is no aspect-level governance architecture within which independent governability is exercised.

The Purposeless Aspect anti-pattern (B3.05) is the failure mode on this side of the pair. Without a coordination purpose, an aspect is a nominal grouping of cells — a label over a collection — rather than a governed coordination arrangement with specified membership, invocation rules, and output integration. Modular cells participating in a purposeless aspect receive no coordination governance: their modularity is architecturally present but coordination-purposeless. The structural virtue B1.08

commits to is without the context that makes it operationally meaningful.

Full co-presence. When B1.04 and B1.08 are jointly present, modular cells with clear operational boundaries participate in aspects through precise coordination rules. Membership is reconfigurable through composition configurability because cells are composable and because the coordination architecture supports the reconfiguration. Conflict handling is tractable because cell operational territories are specified and conflicts between them are identifiable. Invocation rules are precise because cell interfaces are specified. Aspect-level verification is compositional because cells can be independently tested. And cells are modular in a way that is architecturally motivated: their composability, testability, clear interface, and independent governability each serve the specific governance functions the aspect coordination architecture needs them to serve.

7. Detection

Three assessments confirm the pair's operative presence.

B2.36 — Composition configurability assessment. Can aspect membership be reconfigured — can a cell be added to or removed from an aspect — without redesigning remaining member cells? A positive answer confirms both that the coordination governance architecture supports membership change (B1.04) and that cells are independently composable (B1.08). A negative answer localizes to whichever commitment is absent: if the governance architecture does not support reconfiguration, B1.04 is impaired; if reconfiguration requires redesigning co-member cells, B1.08 is impaired.

B2.39 — Modularity verification. Can each cell's behavior be tested independently of its current aspect membership? Can each cell's input and output types be read from its substrate content without examining co-member cell substrates? A positive answer confirms B1.08 is operative. A negative answer — cells whose behavior depends on co-member state, or whose interfaces can only be inferred from the context of a particular aspect — signals the Monolithic Cell failure mode.

B2.19 — Aspect-level inheritance verification. Do the coordination rules per B2.16 function correctly with current cell membership? Can the verification confirm each cell's contribution independently before confirming the coordination arrangement's aggregate behavior? A positive answer confirms both that coordination rules are precisely specified (B1.04) and that cells are independently testable (B1.08). A coordination-rule failure that cannot be localized to a specific cell signals the modularity failure: when cell operational territories are vague, coordination-rule failures are similarly vague in their origin.

Pair completeness check. For each aspect in the architecture: are the coordination rules per B2.16 precisely stated, with invocation rules referencing specific cell input types and output integration rules referencing specific cell output types? Are those precise specifications made possible by the cell modularity B1.08 commits to? A pair that passes this check instantiates the reinforcing relationship this note formalizes. A pair that fails — governance rules that are vague, or modular cells whose modularity

is not exploited by any coordination rules — signals the absence of one or both commitments.

8. Conclusion

The B1.04 + B1.08 composition pair formalizes a reinforcing relationship in which each commitment provides what the other requires in order to operate fully. Aspect coordination governance (B1.04) requires cell modularity (B1.08) because the four governance functions aspects perform — membership reconfiguration, conflict handling, invocation rule specification, and aspect-level verification — are each bounded by the structural properties of cells. Cell modularity (B1.08) is motivated by aspect coordination (B1.04) because the four modularity properties — composability, independent testability, clear interface, and independent governability — each gain operational content from the coordination functions aspects need to perform.

The pair’s operational embodiment is composition configurability (B2.36): membership can be reconfigured because cells are composable and because coordination governance supports the reconfiguration. Multi-aspect participation (B2.18) is possible because cells are independently composable and meaningful because each aspect’s governance of that participation is precise. The failure modes on each side — Monolithic Cell (B3.09) when B1.08 is absent, Purposeless Aspect (B3.05) when B1.04 is absent — name the structural outcomes when one commitment is present without the other.

Subsequent work that adopts the CKS pattern at Paper 2 scope should treat this pair as an architectural unit: aspects and modular cells are not independent design decisions that happen to compose well, but jointly committed architectural properties whose operational meaning is constituted by their mutual dependence.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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